

Programme Code: TU856, TU857, TU858
Module Code: CMPU 3026, CMPU3036

TECHNOLOGICAL UNIVERSITY DUBLIN
CITY CAMPUS, GRANGEGORMAN

TU856 – BSc (Hons) in Computer Science
TU857 – BSc (Hons) in Computer Science (Infrastructure)
TU858 – BSc (Hons) in Computer Science (International)

Year 3

SEMESTER 1 EXAMINATIONS 2025/26

CMPU 3026, CMPU3036 Mobile Software Development

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Instructions To Candidates: Answer **ONLY Four (4)** Questions out of Five
All questions carry equal marks (25 marks)

Exam Duration: 2 hours

Special Instructions /Handouts/ Materials Required: NA

QUESTION 1 (Basics)

Android applications are built from components that work together to manage the app's behaviour, user interface, and configuration.

- (a) Explain the role of the Android Activity Lifecycle in managing app behaviour and user interface components. In your answer, describe the main lifecycle methods and give examples of what typically happens in each.

(10 marks)

- (b) Explain the purpose of the Gradle build system in Android Studio. Give examples of information or settings that are typically defined in a Gradle build file and explain why they are important

(5 marks)

- (c) Explain the role of the AndroidManifest.xml file. List and describe three key elements or declarations commonly included in this file.

(5 marks)

- (d) Describe what happens when a configuration change occurs (for example, device rotation or switching to dark mode). Explain how developers can manage data and user interface state to avoid losing information during such changes.

(5 marks)

[Question One Total Marks 25]

QUESTION 2 (UI and Accessibility)

- (a) Figure 1 represents an Android screen showing a User Form.

Identify and describe the main user interface (UI) elements shown in the screen, and the layout container used to organise them.

(10 marks)

- (b) Explain the purpose of the `strings.xml` file in Android and how it supports flexible, maintainable user interfaces.

(5 marks)

- (c) Define and explain `dp` and `sp` units when defining layout and text sizes in Android.

(5 marks)

- (d) Outline three techniques or features developers can use to make Android apps accessible to all users, including those with disabilities.

(5 marks)

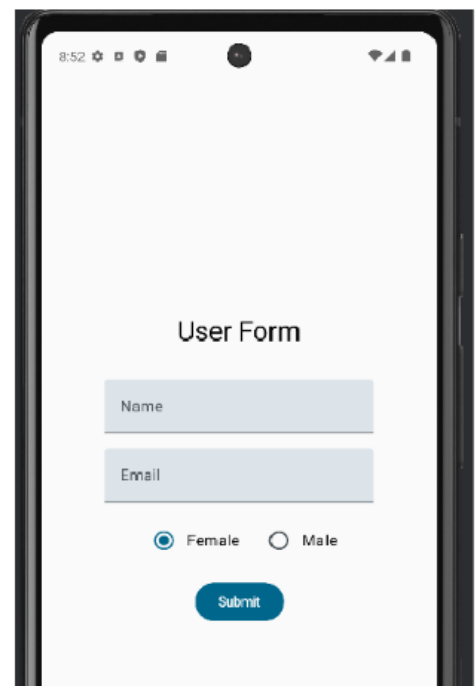


Figure 1 Simple Input Form

[Question Two Total Marks 25]

QUESTION 3 (Room Database)

Explain how data is stored, retrieved, and displayed in an Android application using the Room persistence library and Kotlin coroutines.

In your answer, refer to:

- (a) The purpose of Entities, DAOs, and the Database class (12 marks)
- (b) How coroutines are used to manage database operations (7 marks)
- (c) How the UI observes data changes and updates automatically (6 marks)

[Question Three Total Marks 25]

QUESTION 4 (Networking)

Android apps often need to communicate with web APIs to retrieve and display live data. Libraries such as **Retrofit** and **Volley** are commonly used to simplify this process. Explain how Android manages network communication and data display when using Retrofit or Volley.

In your answer, refer to the following points:

- (a) The main components used in Android network libraries and their roles (10 marks)
- (b) How JSON data is handled and mapped to Kotlin objects (5 marks)
- (c) How asynchronous operations prevent the main thread from blocking. Give an example (5 marks)
- (d) Give two examples of how developers can make the app more reliable when using web data (5 marks)

[Question Four Total Marks 25]

QUESTION 5 (Sensors and Location)

You are developing a mobile fitness application that tracks user movement and orientation. The app must distinguish between walking, running, and standing still, and also determine the direction the user is facing. The application is also using location services to track the user's location.

- (a) Identify and explain the roles of **two different sensors** other than the location services that would be most appropriate for this application. (5 marks)

- (b) At 9:00 a.m., the user is at the initial position (Point A) — no location update is triggered yet.

Location listener is configured with **minTime** = 5 minutes and **minDistance** = 1 kilometre.

Explain the following scenario and indicate whether or not the `onLocationChanged` parameter is triggered during each of the following events:

- At 9:02 am, the user moved 10 kilometres to point B.
- At 9:03 am, the user is returning back to A, which is 5 kilometres away.
- At 9:05 a.m., the user has reached point A again.
- At 9:07 am, the user moved 20 kilometres from point A.
- At 9:10 am, the user moved 40 kilometres from point A.

(8 marks)

- (c) Explain how sensor accuracy and battery consumption might affect the performance of your application and suggest strategies to mitigate these issues.

(7 marks)

- (d) Explain why developers must query the existence of a sensor before using it in an Android application. What method is typically used to perform this check?

(5 marks)

[Question Five Total Marks 25]

****==== End of Exam ====****